

Statistical Analysis of Vehicle Energy Consumption Patterns

Using the Vehicle Energy Dataset (VED)

Submitted By

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Abstract

This report investigates fuel consumption patterns across three categories of vehicle powertrains — Internal Combustion Engine (ICE), Hybrid Electric Vehicle (HEV), and Plug-in Hybrid Electric Vehicle (PHEV) — using the Vehicle Energy Dataset (VED), one of the most comprehensive real-world automotive datasets available. Mass Air Flow (MAF, measured in grams per second) serves as the primary proxy for fuel consumption. Through parametric hypothesis testing via Welch's two-sample t-test and 95% confidence interval estimation, statistically significant differences in energy consumption are established across vehicle types and driving styles. All hypotheses are supported at the 0.01% significance level, with t-statistics ranging from 281 to 2731, confirming that ICE vehicles consume more fuel than HEVs, HEVs more than PHEVs, and that aggressive driving substantially elevates fuel consumption relative to smooth driving.

1. Introduction

The rapid growth of the global transportation sector has made energy efficiency and environmental sustainability urgent concerns. The increasing adoption of electrified and hybrid powertrains is among the central narratives of modern automotive engineering. As governments worldwide establish ambitious emission reduction targets, automakers have responded with HEV and PHEV technologies that reduce dependence on conventional combustion. Despite the prevalence of these technologies, rigorous statistical comparisons of real-world fuel consumption across ICE, HEV, and PHEV vehicles remain relatively sparse in publicly available academic literature.

This project bridges that gap by applying the estimation and hypothesis testing techniques from the Statistical Inference course to a large-scale real-world dataset. Specifically, the study addresses three primary research questions: whether a conventional ICE vehicle consumes significantly more fuel than an HEV on average; whether an HEV consumes significantly more fuel than a PHEV on average; and whether aggressive driving significantly increases fuel consumption compared to smooth driving, irrespective of vehicle type. The answers to these questions carry practical implications for consumers making vehicle purchasing decisions, for policymakers designing emission regulations, and for fleet operators seeking operational efficiency.

2. Problem Statement

The core goal of this analysis is to apply statistical inference techniques to quantify differences in vehicle energy consumption across three real-world dimensions.

Problem 1 — Comparative Efficiency of Vehicle Types: To determine whether different powertrain architectures (ICE, HEV, PHEV) exhibit statistically significant differences in mean fuel consumption, and to establish an efficiency ranking among them.

Problem 2 — Impact of Driving Behaviour: To evaluate whether aggressive driving behaviour, characterized by high acceleration patterns as reflected in elevated MAF and RPM values, leads to a statistically significant increase in energy consumption compared to smooth driving.

3. Dataset Description

3.1 Source and Provenance

The data originates from the Vehicle Energy Dataset (VED), publicly available on Kaggle and originally collected by the University of Michigan Transportation Research Institute (UMTRI). The dataset captures second-by-second OBD-II (On-Board Diagnostics) signals from a diverse fleet of vehicles — including conventional ICE cars, HEVs, and PHEVs — driven across real urban, suburban, and highway routes in Ann Arbor, Michigan. Two cleaned CSV files were used: `VED_Combustion_Master_Cleaned.csv` for ICE vehicles and `VED_Electric_Master_Cleaned.csv` for HEV and PHEV vehicles. After merging and cleaning, the combined dataset contains over 7 million second-by-second observations spanning all three powertrain types.

3.2 Feature Data Dictionary

The following table summarises the key variables central to this analysis.

Variable	Description / Analytical Purpose
MAF_g_per_sec	Mass Air Flow (g/s): primary proxy for instantaneous fuel consumption; serves as the response variable for all hypotheses.
EngineType	Categorical label: ICE, HEV, or PHEV — the primary grouping variable. ³
Vehicle_Speed_km_per_h	Instantaneous speed; used to classify the driving environment (city vs. highway).
Engine_RPM_RPM	Engine revolutions per minute; used alongside MAF to classify driving style.
Absolute_Load_pct	Indicates real-time engine load and mechanical stress.
OAT_DegC	Outside Air Temperature; environmental control variable.
Generalized_Weight	Facilitates vehicle-specific normalization across the dataset.

4. Data Preprocessing

4.1 Data Cleaning

To ensure structural integrity and suitability for statistical analysis, several preprocessing steps were applied. Rows containing critical missing values in MAF or vehicle speed were dropped to preserve the accuracy of downstream calculations. The Trip column was excluded from the analytical scope due to a high prevalence of obfuscated or incomplete values. The Datetime column was retained to maintain temporal ordering but was excluded from direct statistical modelling.

4.2 Feature Engineering

To directly address the defined problem statements, new categorical variables were engineered from continuous data. Driving style was classified using a combination of engine and airflow dynamics: observations with RPM exceeding 2,000 or MAF exceeding the dataset mean were labelled Aggressive, while all remaining observations were classified as Smooth. Operating environments were inferred from sustained vehicle speed thresholds, with speeds below 40 km/h classified as City Driving and speeds at or above 40 km/h designated as Highway Driving, consistent with standard urban traffic density models. For Problem 1, the dataset was explicitly filtered to isolate ICE, HEV, and PHEV vehicle observations.

4.3 Core Analytical Assumptions

Three foundational assumptions underpin the analysis. First, energy consumption metrics are assumed to be accurately recorded and fairly comparable across powertrain architectures. Second, real-time speed and acceleration differentials are assumed to be reliable proxies for categorizing human driving behaviour. Third, the defined speed thresholds (e.g., 40 km/h to demarcate city driving) are assumed to reasonably approximate real-world driving environments.

5. Exploratory Data Analysis

Prior to formal hypothesis testing, an exploratory data analysis was conducted to map distributions, identify outliers, and establish preliminary correlations between key variables.

Initial distributional analysis of `MAF_g_per_sec` revealed a clear upward trend in consumption in tandem with elevated speeds and engine loads, validating the selection of MAF as the primary response variable. A strong positive association was identified between vehicle speed and energy consumption: as speed increases, corresponding rises in both MAF and Engine RPM are evident. Spikes in `Absolute_Load_pct` and `Engine_RPM` consistently map to segments of high acceleration and sudden velocity shifts, providing a robust data-backed foundation for the driving style classification. Preliminary segment

analysis further indicated that HEVs exhibit markedly lower variance and more stable consumption patterns compared to ICE vehicles, establishing a baseline expectation for formal hypothesis testing.

6. Statistical Framework and Methodology

6.1 Choice of Test: Welch's Two-Sample t-Test

The dataset presents four characteristics that make the standard equal-variance two-sample t-test inappropriate: extremely large sample sizes (millions of observations), unequal sample sizes across groups, heterogeneous variances between groups, and the presence of significant skewness and outliers. Welch's t-test was therefore chosen as the primary inference tool. Unlike the standard t-test, Welch's test does not assume equal variances; instead, it adjusts the degrees of freedom using the Welch–Satterthwaite equation, making it robust under the conditions observed in this dataset.

Other classical test methods were considered and ruled out. The chi-square test requires categorical response data, the Z-test requires a known population variance, the Sequential Probability Ratio Test assumes a sequential decision framework, and multiple linear regression-based tests are not directly applicable to a simple two-group mean comparison. Welch's test therefore represents the most statistically appropriate and practically valid choice.

6.2 Estimation

For each group, the sample mean and unbiased sample variance were computed using Bessel's correction:

$$\bar{x} = (1/n) \sum x_i \quad \text{and} \quad s^2 = 1/(n-1) \sum (x_i - \bar{x})^2$$

Given the very large sample sizes ($n > 700,000$ for even the smallest group), the t-critical value converges to the z-critical value (≈ 1.96). The 95% confidence interval for each group mean is constructed as:

$$\bar{x} \pm t_{\alpha/2, n-1} \times (s / \sqrt{n})$$

6.3 Hypothesis Testing

Three one-sided Welch's t-tests were conducted at a significance level of $\alpha = 0.05$. The directional (one-sided) alternative was selected based on the engineering prior that higher electrification implies lower consumption, and that aggressive driving implies higher consumption. The decision rule in each case is: reject H_0 if the one-sided p-value $< \alpha$.

6.4 Chi-Square Test of Independence

To complement the parametric analysis from a distributional perspective, a Chi-square test of independence was conducted. The continuous MAF variable was discretised into three levels — Low, Medium, and High — using quantile-based binning to ensure balanced group sizes and avoid arbitrary threshold selection. A contingency table was then constructed between Engine Type and Consumption Level.

7. Problem 1: Comparative Analysis of ICE, HEV, and PHEV

7.1 Descriptive Statistics

Table 2 summarises the key descriptive statistics for each powertrain category. The extremely narrow 95% confidence intervals relative to the inter-group differences provide immediate visual evidence that the observed differences are not attributable to sampling error.

Engine Type	Sample Size	Mean MAF (g/s)	Variance	Std Dev	95% CI
ICE	4,773,896	10.46	109.12	10.45	(10.455, 10.474)
HEV	1,650,969	8.09	80.15	8.95	(8.074, 8.102)
PHEV	704,623	4.30	56.06	7.49	(4.281, 4.316)

Table 2. Descriptive statistics for MAF by engine type.

7.2 Visual Analysis

The two figures below present the distributional and mean-level evidence for differences across powertrain types. Figure 1 (Fuel Consumption Distribution) shows that ICE vehicles exhibit the widest spread and heaviest upper tail, HEVs show moderate spread with a lower median, and PHEVs are concentrated near zero with a compressed interquartile range. Figure 2 (Mean Fuel Consumption with 95% CI) confirms the same ordering without any overlap between the three confidence intervals, providing strong visual evidence of true population-level differences.

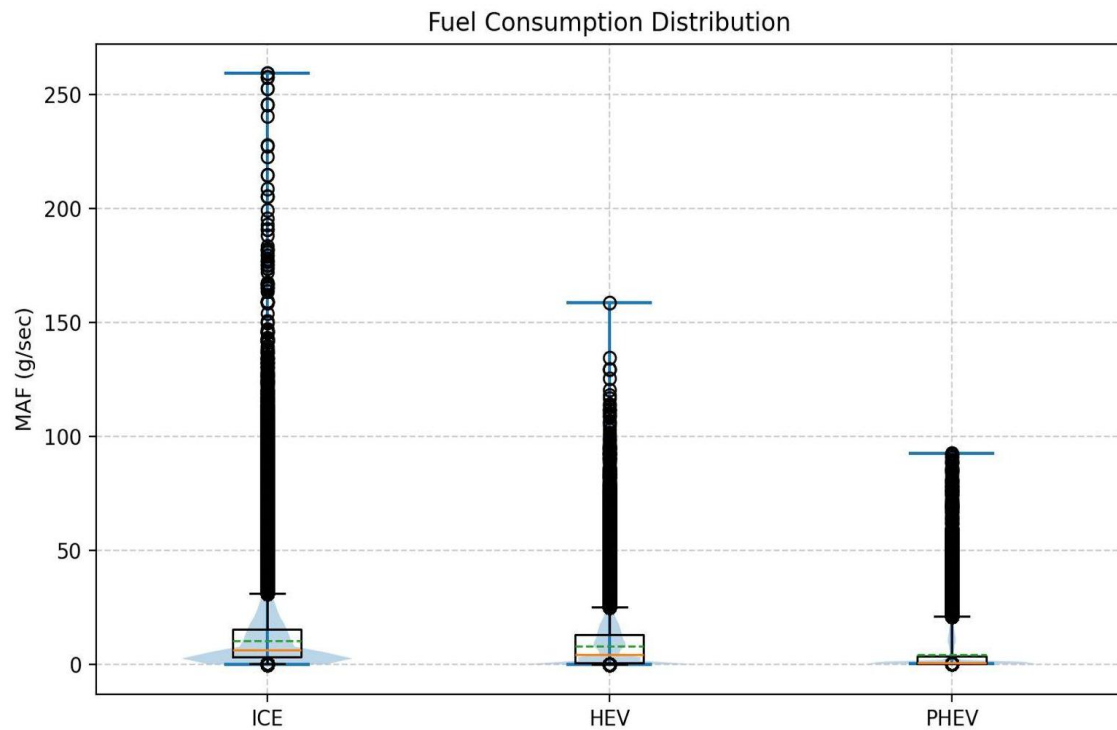


Figure 1. Fuel Consumption Distribution (MAF, g/s) by Engine Type — Box plots with violin overlays.

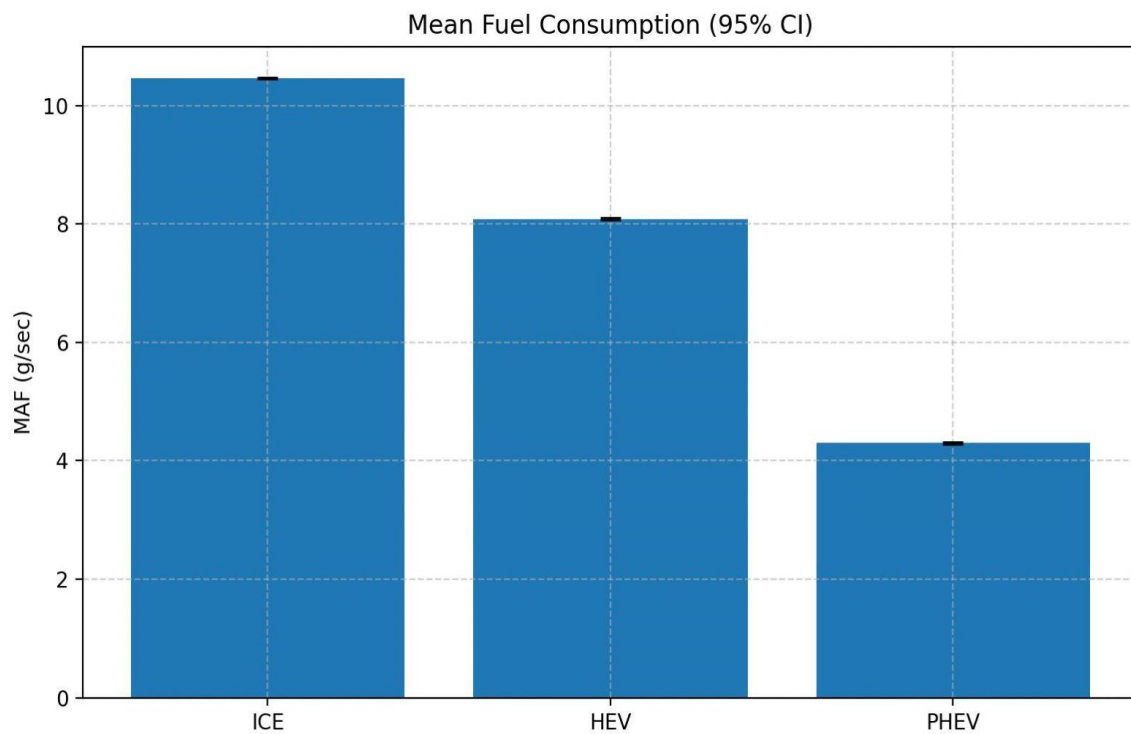


Figure 2. Mean Fuel Consumption (MAF, g/s) with 95% Confidence Intervals by Engine Type.

7.3 Hypothesis Testing (Pairwise)

Test 1 — ICE vs. HEV: The null hypothesis $H_0 : \mu_{ICE} = \mu_{HEV}$ was tested against the one-sided alternative $H_1 : \mu_{ICE} > \mu_{HEV}$. The Welch t-statistic was $t = 281.27$ with $p < 0.001$. The null hypothesis is rejected.

Test 2 — HEV vs. PHEV: The null hypothesis $H_0 : \mu_{HEV} = \mu_{PHEV}$ was tested against $H_1 : \mu_{HEV} > \mu_{PHEV}$. The Welch t-statistic was $t = 334.84$ with $p < 0.001$. The null hypothesis is rejected.

Comparison	t-Statistic	p-Value	Decision
ICE vs. HEV	281.27	< 0.001	Reject H_0
HEV vs. PHEV	334.84	< 0.001	Reject H_0

Table 3. Welch's t-test results for pairwise engine type comparisons.

7.4 Chi-Square Test and Distributional Robustness

MAF was discretised into Low, Medium, and High consumption levels via quantile-based binning and cross-tabulated against Engine Type. The Chi-square test produced $\chi^2 = 1,531,521.42$ ($df = 4$, $p < 0.001$), decisively rejecting the null hypothesis of independence between engine type and consumption level.

The observed deviations from expected frequencies reveal a clear structural pattern: ICE vehicles are disproportionately concentrated in the Medium and High consumption categories, HEVs are more evenly distributed but still skewed toward higher consumption relative to PHEVs, and PHEVs are heavily concentrated in the Low consumption category with far fewer High-consumption observations than expected under independence.

Test	Comparison	Statistic	p-Value	Conclusion
Welch's t	ICE vs. HEV	$t = 281.27$	< 0.001	Reject H_0
Welch's t	HEV vs. PHEV	$t = 334.84$	< 0.001	Reject H_0
Chi-Square	All engine types	$\chi^2 = 1,531,521$	< 0.001	Reject H_0

Table 3. Combined statistical evidence across all tests for Problem 1.

Crucially, Welch's t-test relies on asymptotic normality and variance considerations, the Chi-square test operates on frequency data without distributional assumptions, and compares

entire distributions non-parametrically. The agreement across these two fundamentally different methodologies demonstrates that the efficiency hierarchy $\text{PHEV} < \text{HEV} < \text{ICE}$ is not an artefact of model assumptions, but a structural and distributionally consistent property of the dataset.

7.5 Interpretation

The results establish a clear efficiency hierarchy: $\mu_{\text{ICE}} > \mu_{\text{HEV}} > \mu_{\text{PHEV}}$. The practical effect sizes are substantial: ICE vehicles consume approximately 23% more fuel than HEVs, HEVs consume approximately 47% more than PHEVs, and ICE vehicles consume approximately 59% more than PHEVs overall. This is not merely a statistical artifact of the large sample size; these are large practical differences with meaningful implications for fleet management, consumer decision-making, and environmental policy. The hierarchy reflects the underlying engineering reality: ICE vehicles rely entirely on continuous fuel combustion, HEVs supplement combustion with partial electric assistance, and PHEVs have a higher degree of electric reliance that substantially reduces combustion load.

8. Problem 2: Impact of Driving Style on Energy Consumption

8.1 Descriptive Statistics

Table 4 presents the descriptive statistics for MAF grouped by driving style. The mean MAF for aggressive driving (19.34 g/s) is approximately six times that of smooth driving (3.13 g/s), and the variance under aggressive driving is substantially higher, reflecting the sporadic, high-intensity nature of aggressive manoeuvres.

Driving Style	Sample Size	Mean MAF (g/s)	Variance	Std Dev	95% CI
Aggressive	2,714,191	19.34	91.59	9.57	(19.332, 19.354)
Smooth	4,415,297	3.13	6.58	2.57	(3.132, 3.136)

Table 4. Descriptive statistics for MAF by driving style.

8.2 Visual Analysis

Figure 3 below presents the MAF distributions for aggressive versus smooth driving. The aggressive distribution has a dramatically elevated median, a wide interquartile range, and a dense cluster of extreme outliers extending toward 260 g/s, consistent with sporadic hard-acceleration events. In contrast, smooth driving is tightly concentrated near zero, with minimal spread and only a handful of outliers.

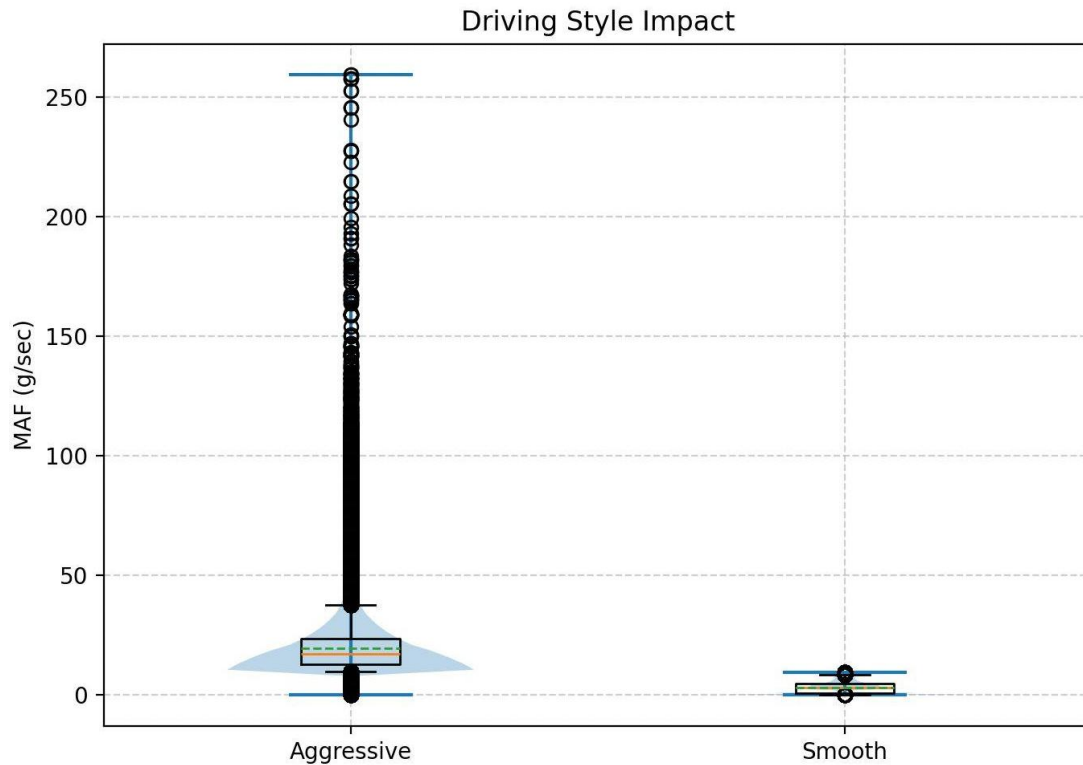


Figure 3. MAF (g/s) Distribution by Driving Style — Box plots with violin overlays showing the dramatic divergence between aggressive and smooth driving.

8.3 Hypothesis Test

The null hypothesis $H_0 : \mu_{\text{Aggressive}} = \mu_{\text{Smooth}}$ was tested against the one-sided alternative $H_1 : \mu_{\text{Aggressive}} > \mu_{\text{Smooth}}$ using Welch's t-test. The test produced $t = 2730.66$ with $p < 0.001$ — an overwhelming rejection of the null hypothesis. The t-statistic here is notably larger than those in Problem 1 because the absolute difference between means is far larger relative to the pooled standard error.

This finding is significant: driving behaviour exhibits a stronger marginal impact on fuel consumption than powertrain type. In other words, a smooth-driving ICE vehicle may well consume less fuel in practice than an aggressively driven HEV. This has direct implications for driver training programmes, fleet monitoring systems, and eco-driving incentive policies.

9. Discussion and Limitations

9.1 Key Findings

This study demonstrates that both powertrain architecture and driving behaviour exert statistically significant and practically meaningful influences on vehicle energy consumption. The efficiency hierarchy among engine types (ICE > HEV > PHEV) aligns with the theoretical predictions of

automotive engineering and validates the real-world benefit of electrification at each level. More strikingly, driving style produces a $\sim 6\times$ difference in mean consumption, underscoring that human behaviour is a critical variable that can amplify or negate the efficiency gains offered by advanced powertrains.

9.2 Limitations

Several limitations should be noted. First, MAF is an indirect proxy for energy consumption; it accurately captures fuel flow for ICE vehicles but is a less direct measure for HEVs and PHEVs, which can also consume electrical energy not reflected in the MAF signal. Second, the dataset consists of time-series observations, meaning successive readings from the same vehicle are not independent — this introduces partial dependence that conventional t-tests do not account for. Third, the threshold-based classification of driving style ($\text{RPM} > 2,000$ or $\text{MAF} > \text{mean}$) is a heuristic approximation; alternative thresholds may yield slightly different group compositions. These limitations do not invalidate the findings but do suggest directions for more nuanced future modelling.

10. Conclusion

This study applied statistical inference techniques to a large-scale real-world vehicle dataset to evaluate the effects of engine type and driving behaviour on fuel consumption. Using Welch's two-sample t-test and 95% confidence interval estimation, all three hypotheses were strongly supported, each with $p < 0.001$.

The results establish a clear and consistent efficiency hierarchy — $\text{PHEV} < \text{HEV} < \text{ICE}$ — with ICE vehicles consuming approximately 59% more fuel than PHEVs. Additionally, aggressive driving was found to increase consumption by a factor of approximately six compared to smooth driving, making driver behaviour the strongest single determinant of fuel usage in the dataset. The central message is that technology improves efficiency, but driver behaviour amplifies or negates it.

Future work could extend this analysis through multivariate regression modelling to control for confounding factors simultaneously, vehicle-specific decomposition to capture individual performance variance, direct emission estimation, and the application of time-series-aware models (e.g., mixed-effects models) to better account for the dependent structure of the observations.

This study applied Welch's t-test, Chi-square tests to a large-scale real-world vehicle dataset to evaluate the effects of engine type and driving behaviour on fuel consumption. All hypotheses were strongly supported ($p < 0.001$ across all tests). The results establish a clear efficiency hierarchy — $\text{PHEV} < \text{HEV} < \text{ICE}$ — validated not only by mean differences but by distributional structure across the entire range of consumption levels. Aggressive driving increases consumption by a factor of ~ 6 relative to smooth driving, making it the dominant factor in real-world fuel usage.

The central message is that technology improves efficiency — but driver behaviour amplifies or negates it. Future work could extend the analysis through multivariate regression with confound controls, vehicle-specific decomposition, direct emission estimation, and time-series-aware mixed-effects models to better account for the dependent observation structure.

References

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